How Much Should Dave Charge for Donuts?

Dave owns Dave's Donut Shop and must make decisions about how much to charge for donuts. As a businessowner he wants to attract a lot of customers as well as make a lot of money (revenue).

The amount of revenue Dave makes can be modeled by the equation $R = (p-1)(200-40p)$, where p represents the price of one donut and R represents the total revenue. **Use as many strategies as you can to figure out the questions below.**

- At what price will Dave make no revenue?
- How much should Dave charge to maximize his revenue?
- What is the most money Dave can make?
- Dave made \$100 from donut sales. How much must he have charged per donut?

Tabular

Price	0	1	2	3	4	5
Revenue	-200	0	120	160	120	0

Analytical

$R=0 \Rightarrow (p-1)=0$ or $(200-40p)=0$

$p=\$1$ or $p=\$5$

Max occurs half-way in between, so at $p=\$3$.

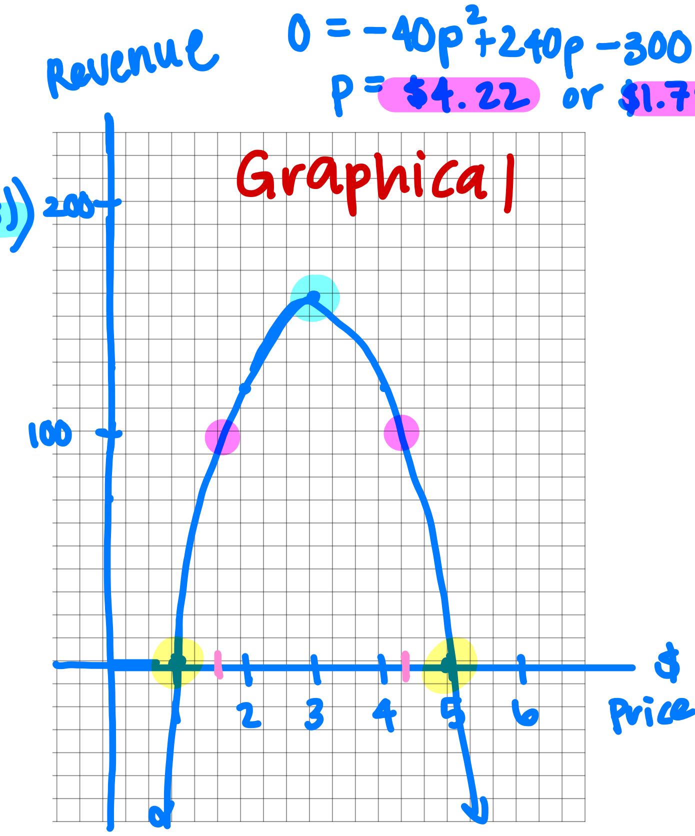
when $p=3$, $R = (3-1)(200-40(3))$

$= \$160.$

$100=(p-1)(200-40p)$

$0=-40p^2+240p-300$

$p=\$4.22$ or $\$1.78$



Dave will break even if he charges \$1 or \$5.

To maximize his revenue he should charge \$3 and he will be able to make \$160.

To make a revenue of \$100 he must have charged \$4.22 or \$1.78

Math Medic

Lesson 0.3 – Solving Equations in Multiple Representations

QuickNotes

Solving equations can be done in multiple representations:

Numerical	Analytical	Graphical
-Make a table, look for values -Plug + chug	-use algebraic methods (factoring, quad. formula)	-Find intersection of 2 curves $\boxed{2^{nd}}$ $\boxed{\text{Trace}}$ $\boxed{\text{5:intersect}}$ y_1 = left side of eqn. y_2 = rightside of eqn.

Check Your Understanding

1. A penny is dropped from the Empire State building. Its height above the ground, in feet, after t seconds is given by $h = 1457 - 16t^2$.
   a. Demonstrate how you can use the table below to *approximate* the time at which the penny landed on the ground.

t	7	8	9	10	11
h	673	433	161	-143	-479

The penny landed on the ground sometime between $t=9$ seconds and $t=10$ seconds because $-143 < 0 < 161$

   b. Find the exact time that the penny landed on the ground, using algebraic methods. Show your work.

$0 = 1457 - 16t^2$

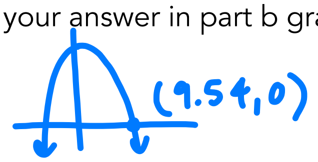
$16t^2 = 1457$

$t^2 = 91.0625$

$t = 9.54$ seconds

Binomial was easy to solve by hand.

   c. Verify your answer in part b graphically.



2. Find the solution(s) of $\sqrt{x+2} = 3x^2 - 4x - 1$ using a graphing calculator. (Remember: keep at least 3 decimal places.)

$x = -0.427$ and $x = 1.864$

3. An up-and-coming artist has 85 followers on Twitter and this number doubles every week. In what week will the artist first reach 1,000 followers? Use at least two different methods to come up with your answer.

Method 1:

Week	0	1	2	3	3.5	3.6
Followers	85	170	340	680	961.7	1030.7

In the 3^{rd} week, ≈ 3.6 weeks

$85(2)^x = 1000$

$2^x = 11.765$

$= 3.556$ weeks